

Sizhe Cheng

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Human-computer interaction and visual analytics for large language models. I build interactive systems that make retrieval-augmented generation, LLM-driven program evolution, and mobile visualization interpretable, steerable, and deployable.

EDUCATION

Nanyang Technological University (NTU)

Master of Engineering by Research, Computer Science Advisor: Prof. Yong Wang

Singapore

09/2025 – Present

Southern University of Science and Technology (SUSTech)

B.Eng. in Computer Science and Technology GPA 3.76/4.0 (90/100)

Shenzhen, China

08/2021 – 06/2025

TOEFL 106 (R28/L30/S20/W28), 2024 | GRE 323 (V155/Q168/AW3.0), 2023

PUBLICATIONS

Six peer-reviewed papers at top-tier venues (CHI, UIST ×2, DIS, TOSEM); two as first author.

1. **Sizhe Cheng**, Songheng Zhang, Dong Ma, Yong Wang. **BAIT: Visual-illusion-inspired Privacy Preservation for Mobile Data Visualization**. [ACM CHI 2026] 
 - Automatically generates a *decoy* chart stacked over a real one so the owner reads the truth up close while a shoulder surfer at a distance is misled; optimizes shape, position, tilt, size, color, and spatial frequency against an explicit model of human perception at varying viewing distance.
 - Validated in two user studies (32-participant controlled lab + 12-participant real-world field study).
2. **Sizhe Cheng**, Jiaping Li, Huanchen Wang, Yuxin Ma. **RAGTrace: Understanding and Refining Retrieval-Generation Dynamics in Retrieval-Augmented Generation**. [ACM UIST 2025] 
 - Built an end-to-end interactive evaluation system that opens the RAG black box: multi-level analysis from aggregate performance down to per-chunk retrieval relevance, generation fidelity, and cross-component interaction; users trace knowledge provenance, localize failure cases, and iterate on retrieval strategy in place.
 - Full-stack implementation (RAG/evaluation pipeline + coordinated D3 visual interface); grounded in a literature review and expert interviews, evaluated through case studies and expert evaluation on real-world deployments.
3. Feng Liang, **Sizhe Cheng**, Yikai Li, Ruijie He, Xiaolin Wen, Yong Wang. **EvoMaestro: Toward Interpretable and Steerable LLM-Driven Program Evolution**. [ACM UIST 2026]
 - Makes LLM-driven program evolution auditable and controllable: Evolution Overview summarizes winning and failed strategies, a novel Radial Tree renders the full evolution history with solution-level comparison, and Maestro Chat explains code in natural language; users merge complementary solutions or steer evolution via natural language.
 - Derived from a formative study with 9 domain experts; evaluated by a real-world use case and a 12-participant study against the state-of-the-art baseline.
4. Can Liu, **Sizhe Cheng**, Feng Liang, Zhibang Jiang, Lingru Huang, Kavinda Athapaththu, Yong Wang. **Proteus: Shapeshifting Desktop Visualizations for Mobile via Multi-level Intelligent Adaptation**. [ACM DIS 2026] 
 - LLM-driven multi-agent system that parses live desktop charts and re-authors them for mobile across a three-level design space: global topology (axis transposition, grid reflow), reference frame (tick decimation, scroll injection), and visual elements (semantic abbreviation, label externalization).
5. Jiongchi Yu, Xiaolin Wen, **Sizhe Cheng**, Xiaofei Xie, Qiang Hu, Yong Wang. **Human in the Loop for Fuzz Testing: Literature Review and the Road Ahead**. [ACM TOSEM 2026] 
 - Surveys human-in-the-loop fuzzing and sets a research agenda across human monitoring, human steering, and human-LLM collaboration, arguing for interpretable visualization and on-the-fly expert intervention in the fuzzing loop.
6. Jian Shen, Minfeng Zhu, Yuxin Ma, **Sizhe Cheng**, Zhizhang Chen, Wei Chen. **KLMVis: Knowledge Graph-Based Retrieval Augmented Language Model Visual Analysis System**. [CAD/Graphics (CADCG) 2024, Nanchang, China]

RESEARCH EXPERIENCE

College of Computing and Data Science, NTUSingapore09/2025 – Present

Graduate ResearcherAdvisor: Prof. Yong Wang

- Lead author on mobile-visualization privacy (BAIT, CHI’26); co-developer on LLM multi-agent systems for visualization adaptation (Proteus, DIS’26) and interactive program evolution (EvoMaestro, UIST’26).
- Responsible for perception modeling and optimization, multi-agent LLM pipelines, and the front-end visual analytics interfaces that ship with each system; ran lab and field user studies (12–32 participants).

Visual Analytics Group, SUSTechShenzhen, China09/2023 – 06/2025

Undergraduate ResearcherAdvisor: Prof. Yuxin Ma

- Designed and built RAGTrace (UIST’25) end to end, from formative interviews through expert evaluation: evaluation metrics, retrieval/generation instrumentation, backend services, and the coordinated multi-view interface.

State Key Laboratory of CAD&CG, Zhejiang UniversityHangzhou, China01/2024 – 09/2024

Research AssistantAdvisors: Prof. Minfeng Zhu, Prof. Wei Chen

- Built interactive tooling for inspecting and improving retrieval-augmented generation systems, including knowledge-graph views in D3.js and the backend query service over a medical (ChatDoctor) knowledge graph, published as KLMVis (CADCG’24).
- Studied vision-language models: fine-tuning for downstream generation quality, and data pruning / condensation to cut training cost at fixed quality; prototyped visual-analytics interfaces to improve VLM interpretability.

TECHNICAL SKILLS

Languages	C, C++, Java, Python, SQL, JavaScript, HTML/CSS
Frameworks & Tools	Vue.js, D3.js, Spring Boot, PyTorch, TensorFlow, Figma, LaTeX, Git, Linux, Android Studio
LLM Engineering	RAG pipeline design and evaluation, LLM multi-agent systems, prompt design, fine-tuning of (vision-)language models, data pruning & condensation
Research Methods	Interactive system design, formative & expert interviews, controlled lab and field user studies, quantitative analysis

AWARDS & COMPETITIONS

- **Silver Medal**, ICPC Asia Hangzhou Regional Contest, *Team Leader*, 2022.
- **Silver Medal**, ICPC Asia Hefei Regional Contest, *Team Leader*, 2022.
- **Outstanding Graduate**, SUSTech, 2025.